



Javier Méndez-Gallego

PhD Student

Date of birth: March 28th, 1999 **Nationality:** Spain

Current Institution: IAA-CSIC (Spain)

Mail address: jmendezgallego@gmail.com

Phone number: (+34) 675 492 025

ORCID:  0009-0006-6222-5813

Work Experience

Dec. 2023 — Now Predoctoral Researcher at Instituto de Astrofísica de Andalucía (IAA-CSIC). Granada, Spain.
Study on the production of galactic cosmic rays in massive young stellar clusters with IACTS and gamma-ray emission produced in star-forming environments.

Relevant publications

Exploring the capability of the HH 80-81 protostellar jet to accelerate relativistic particles.

Méndez-Gallego, J., López-Coto, R., de Oña Wilhelmi, E., et al.
Astronomy & Astrophysics, Volume 695, id. A11. March 2025.
DOI: 10.1051/0004-6361/202452473

Education

Dec. 2023 — Now PhD in Astrophysics at Instituto de Astrofísica de Andalucía (IAA-CSIC) and Universidad de Granada. Granada, Spain.
Thesis title: Cosmic Ray Production and Acceleration in Star Forming Environments and Its Impact in Galaxy Evolution.
Supervisor: Rubén López-Coto *Co-supervisor:* Iván Agudo

Sep. 2022 — Jul. 2023 MSc. in Astrophysics at Universidad Complutense de Madrid. Madrid, Spain. (9.6/10)
Thesis: JWST NIRCIm Imaging Photometric Datasets. Studying the SED of High-Redshift Sources in the Epoch of Reionization. (10/10)
Supervisor: Javier Álvarez-Márquez *Institution:* CAB-INTA

- Jul. 2017 — Sep. 2022* Bachelor's degree in Physics at Universidad Complutense de Madrid. Madrid, Spain. (7.7/10)
Thesis: Development of a stellar-interior numerical model. (9.7/10)
Supervisors: Nicolás Cardiel & Sergio Pascual *Institution:* UCM
- Jul. 2015 — Sep. 2017* Scientific Baccalaureate of Excellence at Instituto Gil y Carrasco. Ponferrada, Spain. (8.8/10)
Thesis: Measuring the displacement of the Eurasian Plate using GNSS data from cities of Castille and Leon.
Supervisors: Enoc Sanz-Ablanedo & Jose M. San Román
Institution: ULE

Collaborations

- Dec. 2023 — Now* CTAO Collaboration. (LST Collaboration)
Contribution: Galactic Physics
- Dec. 2023 — Now* MAGIC Collaboration
Contribution: Galactic Physics

Workshops / schools

- Jun. 2024* CTAO Summer School. Bertinoro (Italy) and La Palma (Spain).
On the theoretical context of the CTAO scientific keys, the telescopes and facilities of the LST-1 prototype, and the software for analyzing CTAO data.
- Feb. 2024* 2nd LST Analysis School. Virtual.
On the analysis of LST data from low level data to post-DL3 analysis. A brief review on the joint MAGIC+LST analysis.
- Jul. 2021* Fermi Summer School 2021. Virtual.
On the analysis of Fermi-LAT data using the Fermi Tools and the Fermipy software.

Conferences and presentations

- Jun. 2025* Studying a protostellar jet as a relativistic particle accelerator: The case of HH 80/81.
Contributed talk at *EAS annual meeting*.
Cork (Ireland)

- Oct. 2024 High-energy gamma-ray emission from a protostellar jet.
Poster at *TOSCA*.
Siena (Italy)
- Jul. 2024 The high-energy spectrum of HH 80-81: Protostellar objects as gamma-ray emitters in the Milky Way.
Contributed talk at *Scientific Meeting of the Spanish Society of Astronomy*.
Granada (Spain)
- Jun. 2024 The high-energy spectrum of HH 80-81.
Student talk at *First CTAO Summer School*.
Bertinoro (Italy)
- May 2024 Cygnus X North: Preliminary analysis results.
Contributed talk at *LST General Meeting*.
Prague (Czech Republic)

Outreach Activities

- 2025 Three contributed talks at StarMus Festival for general public.
La Palma (Spain)
- 2024 — present Collaborator and Secretary of 50 Podcast, a Spanish podcast where we invite different researchers to be interviewed.
Granada (Spain)
- Nov. 2022 “The universe in one room”. An introduction to astronomy during the Week of Science held in UCM.
Madrid (Spain)
- 2020 — 2022 Participation in observational amateur activities held by the university astronomy association of UCM.
Madrid (Spain)
- Jul. 2021 Organizer of an observational lesson in a UCM Summer School.
San Lorenzo del Escorial (Spain)

Other Activities

- Feb. 2025 MAGIC data taking shift as an operator (P274).
La Palma (Spain)
- Nov. 2024 LST Fast Offline Analyzer (P071).
Granada (Spain)
- Mar. 2024 LST Fast Offline Analyzer (P063)
Granada (Spain)